Chemical composition and nutritional value of the most widely appreciated cultivated mushrooms: an inter-species comparative study.

Herein, it was reported and compared the chemical composition and nutritional value of the most consumed species as fresh cultivated mushrooms: Agaricus bisporus (white and brown mushrooms), Pleurotus ostreatus (oyster mushroom), Pleurotus eryngii (King oyster mushroom), Lentinula edodes (Shiitake) and Flammulina velutipes (Golden needle mushroom). Shiitake revealed the highest levels of macronutrients, unless proteins, as also the highest sugars, tocopherols and PUFA levels, and the lowest SFA content. White and brown mushrooms showed similar macronutrients composition, as also similar values of total sugars, MUFA, PUFA and total tocopherols. Oyster and king oyster mushrooms gave the highest MUFA contents with similar contents in PUFA, MUFA and SFA in both samples. They also revealed similar moisture, ash, carbohydrates and energy values. This study contributes to the elaboration of nutritional databases of the most consumed fungi species worldwide, allowing comparison between them. Moreover it was reported that cultivated and the wild samples of the same species have different chemical composition, including sugars, fatty acids and tocopherols profiles.